

# Soumya Tyagi

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## Summary

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Natural Language Processing (NLP), and Generative AI. Proficient in Python, Scikit-learn, TensorFlow, PyTorch, FastAPI, Streamlit, LangChain, and Large Language Models (LLMs). Experienced in building end-to-end AI solutions involving data preprocessing, model training, evaluation, semantic matching, and AI-powered automation. Strong problem-solving skills with a solid foundation in Data Structures and Algorithms (DSA) and core computer science concepts.

## Education

### VIT Bhopal University

2023 – 2027

B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning), **CGPA: 8.43/10**

## Technical Skills

**Programming Languages:** Python, SQL, Java

**AI & Machine Learning:** Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG)

**Frameworks & Libraries:** TensorFlow, PyTorch, Scikit-learn, LangChain, NumPy, Pandas, Streamlit

**Backend & APIs:** FastAPI

**Developer Tools:** GitHub, VS Code, PyCharm, Jupyter Notebook, Google Colab

**Core Computer Science:** Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Database Management Systems (DBMS)

## Experience

### Videonza Technologies (OPC) Private Limited

May 2026 – Jul 2026

#### AI/ML Development Intern

Hybrid

- Contributed to AI/ML development workflows by preprocessing, analyzing, and transforming datasets to support efficient machine learning model development.
- Assisted in training and evaluating machine learning models using **Scikit-learn, TensorFlow, and PyTorch**, improving model accuracy from **82% to 91%** through effective preprocessing and model optimization.
- Performed **data cleaning, feature engineering, feature transformation, and exploratory data analysis (EDA)** using Python and Scikit-learn to improve data quality and identify meaningful patterns.
- Contributed to **model evaluation, validation, and deployment workflows**, helping ensure the reliability, performance, and consistency of AI/ML solutions for practical applications.

## Projects

### AI GitHub Repository Explainer | Python · RAG · ChromaDB · Sentence Transformers · Groq

August 2026

LLaMA · GitHub API

- Developed an **AI-powered GitHub Repository Explainer** using **RAG, FastAPI, Streamlit, ChromaDB, Sentence Transformers, and Groq LLaMA**, enabling developers to query and understand unfamiliar codebases through natural-language questions.
- Implemented semantic code retrieval using 80-line overlapping chunks and vector embeddings, **retrieving the Top-5 to Top-10 most relevant code segments per query** for context-aware LLM responses.
- Engineered an end-to-end RAG pipeline integrating **GitHub ingestion, embedding generation, ChromaDB similarity search, source-aware context construction, and LLM generation** to provide file-level and line-level code references.

### AI Resume Optimizer | Python · Groq (Llama 3.1) · Sentence Transformers · PyMuPDF · ReportLab

June 2026

- Engineered an end-to-end **Generative AI Resume Optimizer** using **FastAPI, Streamlit, Groq LLM, and Sentence Transformers** for resume parsing, ATS analysis, semantic matching, and AI-powered content optimization.
- Improved **job-description keyword alignment by approximately 20%** through semantic similarity analysis and AI-generated resume recommendations.
- Automated resume evaluation, AI feedback generation, interview question generation, and PDF report creation, reducing manual resume review effort by **over 80%**.

## Certifications

- Machine Learning - Udemy
- Applied Machine Learning in Python - Coursera
- Google IT Professional Certificate - Google
- SQL (Intermediate) - Hackerrank